Paper 02 parameter remediation: rerun plan on SDSC Expanse

pmsimstats team

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1. Purpose and scope

This plan details Phase 2 of `docs/70` for paper 02 alone: rerunning every simulation that paper 02 reports under Hendrickson's published response and baseline parameters (58b32a9 `extracted_rp`, `extracted_bp`; Option A of `docs/70` Section 3), on the SDSC Expanse allocation through `simrun --host expanse`. Paper 10 (job 55) shares the generator and is listed as an optional add-on. The companion prognostic paper (jobs 28 and 30) and paper 01 (driver 19) are out of scope here.

The plan assumes Option A. If the CSP 563-calibrated set is chosen instead, only the `--params` value changes; the budget is unaffected.

2. What paper 02 draws on

Paper 02's provenance header and reproducibility section name the following stored runs, all generated by `analysis/scripts/carryover-sensitivity/simulation-core.R` with `default_resp_param()`/`default_baseline_param()` (the sim-core set). Times are from `analysis/remote_runs/*/provenance.yml` and `simrun status` (verified 2026-11-03).

Job	Host	Driver and arguments	Cores	Hours	Core-hours	Feeds
26	zeus	`01-run-factorial.R` (540 cells, 500 reps)	8	5.66	45.2	Decay-form comparison, G1 relative loss (216-cell covariance-moderation slice)
27	zeus	`27-run-tier1-g-vs-f-eq.R --reps 500 --n-boot 1`	8	4.06	32.5	Rungs 1 to 4, reference cell, power by half-life, heatmaps
21	zeus	`06-g-vs-f-comparison.R --reps 500 --n-boot 1`	8	3.06	24.5	Scenario cells (10 of 15 read; see below)
31	zeus	`32-run-g9-gap-fill.R --block A`	6	6.60	39.6	Size grid, CR2 size table
33	zeus	`32-run-g9-gap-fill.R --block A --sub A1-dropout`	8	2.99	23.9	Dropout cells, patient-level convention
32	zeus	`32-run-g9-gap-fill.R --block B`	6	6.80	40.8	Alternative grid
40	mercury	`32-run-g9-gap-fill.R --block C`	1	13.51	81.1	Block C (about 40 zeus core-hours)

Job	Host	Driver and arguments	Cores	Hours	Core-hours	Feeds
34	zeus	`36-run-refree-variants.R --block N`	8	3.53	28.1	Null variants with G10 (also holds $c_{pb} = 0.3$ cells of the companion paper)
35	zeus	`36-run-refree-variants.R --block A`	8	0.71	5.7	Alternative variants with G10
37	zeus	`36-run-refree-variants.R --block S`	8	0.13	1.1	Patient-specific slope
42	zeus	`38-run-e0-isolation.R --reps 1000`	8	3.25	26.0	First-rung isolation; read by the ladder
local	julia	`41-run-g10-backfill.R`	6	n/a	n/a	G10 for jobs 27, 21, 32, 33, 37
local	julia	`diagnose-null-type1-rr2.R`	1	n/a	n/a	Pre-correction size diagnostic (superseded)

Block T (job 36) is not reported and is not rerun. Job 52 (block C on zeus) was cancelled after 118 hours while the same block finished on mercury in 13.5 hours; the cause was not diagnosed (Section 9).

Corrections found during step S0 (2026-11-03, inspected in the sources).

- Block C (`C1-ladder`, `C2-largeN`) is the prognostic-confound ladder. Paper 02's report and summary read only blocks A and B; block C enters only the full-scope ladder of the pre-split paper. It moves with the companion paper and is **not** rerun here.
- Paper 02 still reads the per-occasion dropout cells of job 31 (the heavy-missingness table), as well as job 33's patient-level cells. Block A therefore runs both conventions (`--dropout-per both`); it cannot collapse to one run as first planned.
- The scenario job needs ten cells, not nine: the nine tabulated cells plus `progtvgrow_null`, which the text cites once.

3. Code changes before any submission

All are Phase 0 items of `docs/70`, restricted to what paper 02 needs. Each is committed before submission, because simrun runs the committed tree only.

1. **Published parameters.** Add `published_resp_param()` and `published_baseline_param()` built from `data/extracted_rp.rda` and `data/extracted_bp.rda`; rename the current pair to `simcore2026_*()` and correct the comment that calls them Hendrickson's defaults.
2. **One switch.** `simulation-core.R` reads the set from an environment variable or a `--params published|simcore2026|fixture` argument parsed once, replacing the seven hard-coded calls (lines 436, 1061, 1328, 1415, 1508, 1614 and 1695) and the direct calls in `32-run-g9-gap-fill.R` (line 166), `36-run-referee-variants.R` (line 141) and `38-run-e0-isolation.R` (lines 96 and 206). Every `meta.rds` records the set name and the parameter values.
3. **G10 in the same job.** Add `'E7null_cr2'` to the default `g_specs` of `simulate_cell_combined()`, so jobs 27, 21 and blocks A to C fit G9 and G10 to the same trials. This retires the G10 backfill (`41-run-g10-backfill.R`) and its pairing guard.
4. **Trim unreported work.**
   - `01-run-factorial.R --arch mvn` to run only the covariance- moderation cells the paper uses.
   - `06-g-vs-f-comparison.R --cells paper02` to run only the ten cells the paper reads.
   - `32-run-g9-gap-fill.R --dropout-per both` runs the per-occasion and patient-level dropout cells in one job.
5. **Separate output directories.** Write to `analysis/data/carryover-sensitivity/published/<family` leaving the sim-core outputs in place for the provenance record and any sensitivity comparison. `guard_out_dir()` refuses to write into a directory, or resume from its checkpoints, under a different set.
6. **Checks.** `42-check-param-sets.R` passes under the published set before anything is submitted.

Status of Section 3 (2026-11-03)

Done, uncommitted:

- `simulation-core.R`: `published_*()`, `simcore2026_*()` and `fixture_*()` sets; `default_*()` dispatch on `--params` or `PMSIM_PARAMS` (default `simcore2026`); `param_meta()`; `guard_out_dir()`; G10 in the default `g_specs` of `simulate_cell_combined()`.
- **The analysis puts the biomarker on the SD scale.** `prepare_long_data()` centers `bm` at the set's mean and divides by its SD. The published biomarker has mean 124.33 and SD 15.36, while the analysis code and the coverage truth ($\theta = -c_{bm}\sigma_{BR}$) assume SD units. The map is affine, so tests of `bm:Dbc` are unchanged, and under `simcore2026` it is the identity.
- Drivers 01, 06, 27, 32, 36 and 38: the guard, `params` in `meta.rds`, and the trimming flags (01 also gains `--out-dir`, since it wrote to a fixed path; 06 gains a `meta.rds`).
- `42-check-param-sets.R`: 14 checks, all passing locally (published values equal 58b32a9; dispatch; SD of change from baseline 19.02 and 16.01 against 19.18 and 16.03 implied by Σ; analysis `bm` mean -0.001, SD 1.007; E9 power 0.280 against the closed form 0.285, 1,000 replicates; guard behavior).
- Regression (verified locally): under `--params simcore2026` the edited core reproduces the committed core's estimates, standard errors, p-values and selected half-lives exactly on a Hybrid alternative cell and a MAR dropout cell; the only difference is the added G10 row.

Not done: the package-level `published_resp_param()` export of `docs/70` Phase 0 item 1 (paper 02's drivers do not load the package), and the parameter table in NOTATION.md and CLAUDE.md.

Default set. The default stays `simcore2026`. About fifty scripts source `simulation-core.R`: paper 01's driver 19, paper 10's `paper10-common.R`, the paper 11 and paper 12 drivers, the closed-form checks, and the check scripts 35, 39 and 41. Most of them resume into stored output directories that have no guard. A `published` default would have switched all of them without notice (a first draft of step S0 did this; corrected on review). Every production command in Section 7 therefore passes `--params published` explicitly, and the S5 checks confirm that each `meta.rds` records `published`.

Verified on review (2026-11-03). The `--params` flag selects the set, overrides `PMSIM_PARAMS`, rejects an unknown name, and the guard refuses a directory written under another set (driver 32, smoke scale). G10 rows are present in the gap-fill output, and driver 32's printed summary now lists them. `dropout_per = 'occasion'` is the core default that job 31's cells used, so the restored per-occasion cells match job 31's convention. Driver 01 filters its grid before `furrr` assigns its per-element seed streams, so the `--arch mvn` run does not reproduce

job 26's draws for the same cells. This does not matter for a new published-set run, but the run is not paired with job 26.

4. Expanse readiness (no project job has run there yet)

Infrastructure was verified piece by piece on 2026-10-26 (vignette, `~/dotfiles/share/simrun/vignette.md`, 'Using simrun on Expanse'), and `simrun-server-slurm` already passes `--nodes=1`, `--constraint=lustre`, `--cpus-per-task` and the `sds154` account, and the job wrapper exports `SIM_CORES`. No project job has run end to end: no `analysis/remote_runs/*-expanse` directory exists.

1. **Session login.** `ssh expanse true` by hand, to answer the TOTP prompt; the multiplexed connection lasts four hours, so long monitoring sessions need a fresh login.
2. **Remaining balance.** `expanse-client user -r expanse` on the login node. The trial allocation is 1000 core-hours, a hard cap, through about 2027-04-20 (six months from 2026-10-20).
3. **Image.** No current local image exists (the only one is `ghcr.io/rgt47/pmsimstats-ng:b9c4bbd`). Run `make docker-build` (default `PLATFORM = linux/amd64`, which Expanse needs) at the submission commit; simrun checks the hash label against `Dockerfile` plus `renv.lock` before shipping. The image is about 5.5 GB, so the first `--ship` takes a while.
4. **Memory charging (to verify).** On Expanse's `shared` partition a job is believed to be charged by whichever is larger, its cores or its memory as a share of a node (about 2 GB per core). Requesting 24 GB for 8 cores, as on zeus, would then be charged as 12 cores. Confirm in the Expanse user guide and size `--mem` from the measured peak (step S2) rather than from the zeus settings.

5. Budget

Assumptions (not yet measured).

- Expanse per core runs at about half of julia's speed (verified with the base-R benchmark, 2026-10-26). Zeus's per-core speed relative to julia has not been measured; the estimates below take zeus as equal to julia, so Expanse costs **twice** the zeus core-hours. Mercury is taken as half of zeus (config note).
- Fitting G10 inside each combined job adds about **20 percent** to that job (one more AIC grid and refit per replicate). Not measured.
- The noisier published parameters do not materially change fit time or convergence. Plausible, not checked.

Revised 2026-11-03 after the corrections in Section 2.

Item	Zeus core-hours	Expanse core-hours (x2)
Factorial, covariance-moderation slice only (216 of 540 cells)	18	36
Job 27 grid + G10	39	78
Scenario cells, 10 of 15 + G10	20	39
Block A, `A1-dropout`, both dropout conventions + G10	58	115
Block A, `A2-size` cells + G10	19	38
Block B + G10	49	98
Referee N, A, S	35	70
E0 isolation	26	52
Paper 02 total	**264**	**526**
Smoke and calibration (S1, S2)		15
Optional: paper 10 (job 55)	14	29
Untrimmed alternative (full factorial, all 15 scenario cells)	300	600

Block A is split by sub-block from the job 31 and 33 times (`A1-dropout` at about 24 zeus core-hours per 12,000 cell-replicates, `A2-size` at about 16), which keeps every job under eight hours. The trimmed run plus calibration is about **540 core-hours**, a little over half the allocation, leaving about 450 for failures, one rerun of the largest job and paper 10. The 2x factor is the largest uncertainty; step S2 measures it and step S3 is a hard go/no-go gate on the result.

The zeus alternative. The same work took about 45 hours of wall-clock time on zeus, run serially, at no allocation cost. Expanse buys concurrency: every job can run at once, so the whole set takes roughly one working day of wall-clock time instead of two to three days, and zeus stays free for paper 01's driver 19 and the Phase 3 papers. That trade is sound only if those zeus jobs are actually queued; otherwise zeus alone is the cheaper choice. A split (the large blocks on Expanse, the rest on zeus) is allowed, subject to the pairing rule in Section 6.

6. Pairing and platform rule

Results are not draw-for-draw identical across platforms (different CPUs and BLAS kernels, even with the same image).

- Within a job, every specification is fitted to the same trials, so within-job paired comparisons (G9 against G10, the ladder rungs on job 27) are safe on any host.
- **Job 27 and the E0 isolation (job 42) must run on the same host**: `39-check-e0-isolation.R` reproduces E0 and G1 of job 27 on a subset, and the ladder reads both.
- **Blocks A and B** are assembled into one summary by `33-assemble-gap-fill.R`; run them on one host and one commit so the table has a single provenance.
- The referee-variant blocks have their own seeds per cell and are unpaired by design; they may run anywhere.
- All production jobs run at one commit and one image hash. Any commit after the first production submission that touches `simulation-core.R` invalidates the jobs already submitted.

7. Execution steps

S0. Prerequisites (local, no allocation)

1. Code changes of Section 3, committed; the moment check passes; tinytest passes.
2. Local smoke for each driver under `--params published` at two or three replicates (`01-run-factorial.R --smoke`, the `--dev` modes where they exist), so failures surface before Expanse is charged.
3. `make docker-build`; confirm the image's architecture is amd64 and its `zzcollab.dockerfile.hash` label matches the commit.
4. Section 4 checks: login, balance, `SIMRUN_ROOT` on Lustre project storage.

S1. Smoke job on Expanse (about 1 core-hour)

```
simrun --host expanse submit --ship --cores 4 --mem 8 \
  --time 00:30:00 --label p02-smoke -- \
  Rscript analysis/scripts/carryover-sensitivity/01-run-factorial.R \
  --smoke --params published
```

Pass criteria: the `.sif` builds; the job reaches `running` and `done` with exit 0; `simrun --host expanse fetch` lands `analysis/remote_runs/job-N-<sha>-expanse/` with `provenance.yml` recording host, partition and image digest.

S2. Calibration job (about 12 core-hours)

Rerun stored job 35 (referee block A: 0.71 hours on 8 zeus cores) under the **sim-core** set, same command and seed, at 16 cores:

```
simrun --host expanse submit --cores 16 --mem 32 --time 02:00:00 \
  --label p02-calib-refA -- \
  Rscript analysis/scripts/carryover-sensitivity/36-run-referee-variants.R \
  --block A --cores 16 --params simcore2026 \
  --out-dir analysis/data/carryover-sensitivity/expanse-calib/block-A
```

It measures three things:

1. **Speed ratio** r = Expanse core-hours / 5.7 zeus core-hours.
2. **Peak memory per worker** (`sacct -j <id> --format=MaxRSS,ReqMem` on the login node), which sets `--mem` for production.
3. **Cross-platform reproducibility**: whether the per-replicate estimates equal job 35's. If they agree to numerical tolerance, the pairing rule of Section 6 can be relaxed; if not, it stands.

S3. Go/no-go gate

Projected cost $= r \times 264$ zeus core-hours, with the 1.2 G10 allowance replaced by the measured overhead. Proceed on Expanse if the projection plus 25 percent reserve fits the remaining balance. Otherwise move job 27 plus the E0 isolation (one pair, about 65 zeus core-hours) and the factorial to zeus, keep blocks A and B and the referee blocks on Expanse, and recompute.

S4. Production submissions

Sixteen cores per job (the benchmark showed efficiency falling at 60 and 120 cores; 16 keeps fork overhead and memory contention low), `--mem` from S2, `--time` at 1.5 times the projected wall-clock time so a hung job (compare zeus job 52) cannot drain the allocation. Projected wall-clock times assume $r = 2$.

Label	Command (all with `--params published --cores 16`)	Projected hours	`--time`
p02-factorial	`01-run-factorial.R --arch mvn`	2.3	04:00:00

Label	Command (all with `--params published --cores 16`)	Projected hours	`--time`
p02-tier1	`27-run-tier1-g-vs-f-e0.R --reps 500 --n-boot 1`	49	08:00:00
p02-e0iso	`38-run-e0-isolation.R --reps 1000`	33	05:00:00
p02-scen	`06-g-vs-f-comparison.R --reps 500 --n-boot 1 --cells paper02`	24	04:00:00
p02-gf-dropout	`32-run-g9-gap-fill.R --block A --sub A1-dropout --dropout-per both`	72	11:00:00
p02-gf-size	`32-run-g9-gap-fill.R --block A --sub A2-size`	24	04:00:00
p02-gf-B	`32-run-g9-gap-fill.R --block B`	61	10:00:00
p02-ref-N	`36-run-referee-variants.R --block N`	35	06:00:00
p02-ref-A	`36-run-referee-variants.R --block A`	07	01:30:00
p02-ref-S	`36-run-referee-variants.R --block S`	02	00:30:00

Each command also takes `--out-dir` under `analysis/data/carryover-sensitivity/published/`. Submit in two waves: first the factorial, `ref-A` and `ref-S` (small; they confirm the production settings), then the rest together. With every job running at once, the set finishes in about six to ten hours of wall-clock time plus queue waits. Monitor with `simrun --host expanse status`; the four-hour login window means re-authenticating to fetch.

S5. Fetch, apply and check each job

1. `simrun --host expanse fetch N`, then `apply` into the `published/` tree.
2. Per job: pre-flight eigenvalues all positive (expected, because the correlation matrix is unchanged; inferred, not verified), non-convergence rates

beside the sim-core run's, null-cell rejection rates inside the two-MCSE band, `meta.rds` records `params = published`.
3. `39-check-e0-isolation.R` against the new job 27 output.
4. `33-assemble-gap-fill.R` repointed to the new job directories (Expanse job numbers are separate from zeus's and carry the `-expanse` suffix).

8. Downstream regeneration and text

1. Remove the G10 backfill block from `report.Rmd` and `summary.Rmd` (`bf_sum`, `bf_all`, `g21b`, `sgf_g10`) and read G10 from the main summaries; update the pairing sentence (54,989 of 55,000) since the guard no longer exists.
2. Repoint every `readRDS()` path to `published/`; rerun `34-referee-reanalysis.R`, `37-ladder-tables-figures.R --scope predictive`, `40-stage-power-figure.R`, `12-hendrickson-heatmap.R` and `02-summarise-grid.R`.
3. Methods: state the response and baseline parameter table and that it equals paper 01's (the referee's M1 point disappears for paper 02 once this is true).
4. Reproducibility section: host, partition, commit, image digest, job numbers, cores and times, from the new `provenance.yml` files.
5. Rewrite the Results from the new numbers. Power falls everywhere (40 to 70 percent more noise SD); the ranking of specifications probably survives (inferred), but size claims under misspecification and the adequate-power boundary may move.
6. `perl tools/notation-lint.pl`, then `bash tools/render.sh` for `report.Rmd` and `summary.Rmd`.

9. Risks

- **Budget.** The 2x factor and the 20 percent G10 overhead are assumptions; S2 and S3 exist to bound them before most of the allocation is spent.
- **Hung jobs.** Zeus job 52 ran 118 hours on a block that took 13.5 hours elsewhere. The cause is unknown; `--time` limits cap the loss on Expanse, but a repeat would cost a resubmission.
- **Login window.** TOTP every four hours makes unattended monitoring impossible; fetches wait for the next login.
- **Queue waits.** The `shared` partition is shared with other users; sixteen-core jobs normally start quickly, but this is not guaranteed.
- **Scope creep.** The companion prognostic paper (jobs 28 and 30, about 69 zeus core-hours) carries the same parameter defect and would roughly exhaust the remaining reserve; plan it separately.

10. Evidence status

- **Verified:** job commands, cores and times (provenance files and `simrun status`); Expanse per-core speed against julia (vignette benchmark); `simrun-server-slurm` options; that no Expanse project job exists; the local image inventory.
- **Inspected:** the paper 02 source-to-job mapping (report header and reproducibility section); the parameter call sites listed in Section 3; that only `36-run-referee-variants.R` already fits G10.
- **Assumed:** zeus per-core speed equal to julia's; G10 overhead 20 percent; cost proportional to cell count when trimming; Expanse memory-based charging; unchanged fit time under the published set.
- **Not checked:** the remaining allocation balance; whether paper 02 still narrates the per-occasion dropout cells; the cause of zeus job 52.